

The real cost of credit constraints: Evidence from micro-finance

Renuka Sane Susan Thomas*

June 2014

Abstract

This paper studies the effect of a law passed by the Indian state of Andhra Pradesh in December 2010 that effectively banned micro-credit lending. The analysis uses quarterly data on average household consumption from 2008 to 2013, for 200 regions in India. Regions in Andhra Pradesh are matched to regions that did not face the ban. A difference-in-difference estimation of changes in these matched regions are used to establish causal impact. The results show that the average household consumption in the state dropped by 17.3 percent and that consumption became more volatile, suggesting that a ban is not an optimal intervention to improve customer welfare.

JEL classification: D14, G21, G28

Keywords: micro-credit, consumption smoothing, household finance, regulatory ban, natural experiment

*Email: renuka@saner.org.in, susant@igidr.ac.in, URL: <http://www.ifrogs.org> We thank Ajay Shah for useful inputs into the research design. We also thank the participants at the IGIDR seminar series, the ISI Delhi seminar series and the Emerging Markets Finance 2012 conference for helpful suggestions. The views expressed in this paper belong to the authors and not their employer.

Contents

1	Introduction	3
2	The research setting	5
2.1	A natural experiment: the micro-finance ban in Andhra Pradesh	6
2.2	Research questions	7
3	Data characteristics	7
3.1	Income categories	8
3.2	Consumption	9
3.3	Household borrowings	9
4	Research design	10
4.1	Matching procedure	10
4.2	The difference-in-difference (DID) estimator	13
4.3	Evaluating the design	14
4.3.1	Is there match balance?	14
4.3.2	Analysing a placebo	16
4.3.3	What is the size of the impact that can be resolved by the design?	16
5	Results	17
5.1	Did consumption fall when access to micro-credit was reduced?	17
5.2	Did volatility of consumption rise with lower access to micro- finance?	21
5.3	Is there an effect on average saving?	21
5.4	Which groups were more affected?	22
6	Threats to validity	23
6.1	Did another event in AP cause the results?	24
6.2	Are the results sensitive to the matching strategy?	26
7	Conclusion	27
	Appendix	32
A	The CES data	32
B	List of districts in the control	32

1 Introduction

The proliferation of micro-credit, in the form of payday lending and micro-finance, and the rise in over-indebtedness of low-income households has led to questions about the extent to which micro-credit is welfare enhancing (Melzer, 2011). The for-profit nature of micro-finance firms, and their alleged predatory lending practices and usurious interest rates have also raised regulatory concerns.

In some cases, regulators have intervened to restrict access of households to such credit. A literature has developed around using these regulatory interventions as natural experiments to analyse the impact of restricting credit access to pay-day lending. It finds that restricting payday lending leads to a reduction in household welfare (Morgan and Strain, 2008; Zinman, 2010; Morse, 2011). These add to the literature on the welfare impact of micro-finance by both going beyond the scope of microevaluation using the traditional RCT approach, and by analysing the impact of credit withdrawal rather than increased access. However, a limitation is that they do not observe household consumption directly. Instead, they rely on proxies of household welfare such as bounced cheque payments, foreclosures on mortgages, and small property crimes.

This paper analyses the impact on average household consumption of a policy intervention that forced a closure of the micro-finance industry in the Indian state of Andhra Pradesh (AP). This is an ideal natural experiment to study the effect of credit withdrawal in three respects. First, the ban was unexpected and complete. Second, it was an intervention on a large scale: AP had a population of 84 million in 2010 which exceeded that of Germany at that time. An estimated 27 million of the population were in households that borrowed from the micro-finance industry (Srinivasan, 2012).

Lastly, the paper uses a high quality panel data with average household consumption within 200 regions across all states in India, observed for every quarter starting from 2008. This allows the estimation of the change in household welfare in the regions with the ban on micro-credit in 2010. It also allows the identification of control regions such that they are economically *matched* to regions in AP, but where the state government did not implement a similar ban on micro-finance. The causal effect of the ban is then calculated using a difference-in-difference estimation of the average household consumption of the change the AP regions and the matched regions in the periods before and after the ban.

The analysis finds that there is a significant decline in average household consumption of households in the AP regions, which dropped by 19.5 percent compared to the control. One explanation for the magnitude of the drop could be the high indebtedness in the state: the number of borrower households in AP is significantly high at 85 percent compared to 37 percent of borrower households in the rest of the country. But even after adjusting the selection of the controls appropriately, the estimated impact of the ban is a significant drop in average household consumption of 17.3 percent compared to controls. Additionally, there is evidence that the volatility of average consumption rose, which suggests greater difficulties by households in AP to smooth consumption. The impact was larger in rural regions, which typically have access to fewer sources of credit.

The large negative impact on average consumption in the AP regions suggests that the ban affected all households in the region rather than just the low income households for whom micro-finance is the more significant source of credit. In this, the episode being analysed is reminiscent of the economy-wide effects in episodes of deleveraging by a subset of the economy (Eggertsson and Krugman, 2012). The consequences of a large-scale credit withdrawal may not be limited to direct borrowers of micro-finance institutions. Instead, there may be ramifications on the economy at large owing to the interconnections between the borrowers and the rest of the economy.

This paper contributes to the literature about outcomes of household consumption when access to credit is abruptly withdrawn. To our knowledge, it is the first rigorous analysis of a remarkable natural experiment of a state-wide ban imposed on the micro-credit industry in India. The motivation for the intervention was consumer protection. But such an extreme policy intervention adversely affected the entire state. While regulatory interventions in the name of consumer protection can be welfare-improving, a ban is not.

The paper starts with a discussion on the role of credit access in household consumption in Section 2, followed by a description of the ban on micro-finance in AP in Section 2.1. The research hypotheses to be tested are described in Section 2.2. Section 3 describes the data, while the research design used is presented in Section 4. Section 5 discusses the findings, and Section 6 tests the robustness of these results. Section 7 concludes.

2 The research setting

Even though the theoretical literature on how credit access affects household consumption is rich and old, there is relatively sparse empirical research on changes in credit access to households. A limitation is the paucity of databases on the household balance sheet (Campbell, 2006). Most empirical research, therefore, relies on proxies of household welfare. One of the branches of this literature is the research that uses regulatory restrictions on pay-day lending across states in the U.S. as natural experiments. Here, Morgan and Strain (2008) analyse patterns in household default data such as bounced cheques, complaints against debtors and bankruptcy filings, Zinman (2010) uses employment status and Morse (2011) use foreclosures on mortgages and small property crimes.

Direct evidence on the response of household consumption comes from the growing literature based on randomised control trials (RCTs) (Banerjee, Duflo, Glennerster, and Kinnan, 2013; Augsburg et. al., 2012; Crépon, Devoto, Duflo, and Parienté, 2011; Karlan and Zinman, 2011, 2010). However, much of these are micro-evaluations that observe the impact of increasing credit access only after the credit intervention. There is an emerging literature that evaluates the general equilibrium effects of increased micro-credit access (Kaboski and Townsend, 2012). However, this literature offers no direct evidence on the effect of credit withdrawals.

A new strand of research on household consumption and credit supply has risen in the aftermath of the 2008 financial crisis. For example, Mian and Sufi (2010) argue that frictions in the supply of credit make it difficult for highly leveraged households to continue to make durable consumption purchases and residential investments during the recession of 2007-2009. Households with high exposure to micro-credit are likely to be similar to the highly leveraged households at the time of the financial crisis.

In this paper, we use a natural experiment of an unexpected and effective withdrawal of micro-credit access at the level of a large state in India, and data with quarterly observations on average household consumption available both before and after the intervention. This overcomes the limitation of lack of data availability to a certain extent, and directly address the question of what effect large scale micro-credit withdrawal has on household welfare.

2.1 A natural experiment: the micro-finance ban in Andhra Pradesh

The state of Andhra Pradesh (AP) was the locus of growth of the micro-finance industry in India. It was at the forefront of promoting the Self-Help Group (SHG) programs, which was the state-promoted version of a joint liability loan to groups of women through the largely publicly owned banking sector. It was also the state where the largest micro-finance institutions (MFIs) were head-quartered. AP also had the largest share of micro-credit borrowers and loans outstanding by value in India. At the same time, there were also many instances when MFIs were accused of mis-selling and using coercive collection practices.¹

The first episode of the State threatening action against the micro-finance industry was in 2005. Despite the closure of one MFI and warnings issued at the time, the government claimed that MFIs continued to engage in irresponsible lending practices.² In response, on October 15th 2010, the state government proposed an ordinance that imposed operational constraints on the MFIs,³ which was enacted as law in December 2010. The law aimed to “protect the women of Self Help Groups from exploitation by the Micro Finance Institutions in the State of Andhra Pradesh”, and severely restricted the collection of loan repayments from customers, or the origination of new loans. Any minor annoyance to customers could now justify court proceedings. As a consequence, more than Rs.7000 crore worth of loans in AP effectively went into default, with recovery at about 10% (Srinivasan, 2012). Loan disbursements between September 2010 and March 2011 were 1.7% of loans disbursed in the previous six months. Outside AP, assets of the micro-finance industry rose by 25% during the same period (MFIN, 2012).

The AP state government is said to have enabled the disbursal of loans through the SHGs to compensate for the closure of the micro-finance industry. Provisional data indicates, however, that the number of SHGs availing loans during the year declined by 0.18 million and disbursements declined

¹There are several sources detailing the role played by AP in the growth of the micro-finance industry in India, and the subsequent problems that bedeviled the industry in this state. These include Datta and Mahajan (2003), Srinivasan (2010), Shylendra (2006), Sa-dhan (2007), Arunachalam (2010), Sane and Thomas (2013).

²Such practices included lending without due diligence on the ability of the group to repay, lending for consumption and consumer durables, lack of transparency in operations, charging usurious interest rates, and employing coercive recovery practices (Yerramilli, 2013).

³State government of Andhra Pradesh (2010)

by Rs.3.5 billion compared to the previous year. The shortfall of credit to households was estimated to be about Rs.30 billion (Srinivasan, 2012).

2.2 Research questions

We use this episode as a natural experiment to answer the following questions about how micro-credit withdrawal affects consumption:

1. Does average household consumption get affected when access to micro-finance is reduced, even when micro-credit appears to be a small component of a larger credit industry?
2. Does the volatility of average consumption change?
3. Is there an effect on average saving?
4. Are regions with greater micro-finance exposure more affected?

At the time of the ban, borrowers received a windfall from non-repayment to MFIs. To the extent that households are forward-looking, the fall in consumption may be tempered as the money gets consumed over time.

3 Data characteristics

The database used in the analysis is the “Consumer Pyramids” (CP).⁴ The data consists of quarterly averages calculated for the households within a defined geographical region called a *Homogeneous Region* or HR.⁵ The database covers a total of 200 HRs in India and 14 HRs in the state of AP.⁶ Among the HRs in AP, there are seven in a rural region and seven in an urban region, which provides variation in levels of credit constraints as measured by credit access. Typically, rural regions have a significantly lower level of formal financial presence compared with urban regions.

⁴The CP database is created by the Centre for Monitoring Indian Economy (CMIE). <http://www.consumer-pyramids.com>. It is based on a national panel survey covering 150,000 households, which has been administered every quarter from June 2009 onwards.

The standard data-set in India on consumption expenditure is the Consumption Expenditure Survey (CES), carried out by the National Sample Survey Organisation (NSSO). More details on the NSSO and reasons for not using it are presented in the Appendix.

⁵Each HR is a group of two or more districts within a state.

⁶Each region will cover different number of households. For example, the 14 HRs in AP covers around 7% of the survey households.

Table 1 Income categories for all India

Income groups are formed at various percentiles, with the corresponding income values rounded to the nearest thousand rupees to reflect how respondents report their incomes.

(in September 2010)			
	Annual household income (Rs.)		% share in
	Lower limit	Upper limit	total sample
I-1	1,000,000	Infinity	1.0
I-2	720,000	1,000,000	1.4
I-3	360,000	720,000	8.8
I-4	240,000	360,000	11.7
I-5	180,000	240,000	10.9
I-6	120,000	180,000	16.6
I-7	96,000	120,000	9.9
I-8	60,000	96,000	19.3
I-9	36,000	60,000	15.2
I-10	24,000	36,000	3.5
I-11	0	24,000	1.7

We use two sets of information to understand the household response to a ban on micro-credit:

1. Aggregate household income and consumption expenditure, as well as the number of households, which are reported each quarter.
2. Financial participation such as average borrowings in an HR is reported as fraction of households in the region that borrow and borrow from specific sources such as friends and family, money-lender, bank and micro-finance (MFI/SHG).

We next describe the average household income, consumption and financial participation, for India and for the HRs in AP from the database.

3.1 Income categories

Table 1 reports the various income categories reported in the CP database and the fraction of the sample that fall into these income categories. For instance, the largest fraction of the sample fall in the “I-8” category (nearly 20%), where the annual income of the household is between Rs.60,000 (USD 1111) and Rs.96,000 (USD 1778). There is limited representation of categories with either very high incomes or very low incomes in the sample surveyed in the database.

Table 2 Components of consumption

The table presents components of average household consumption reported in the CP database for AP in September 2010 which is two quarters before the ban, and the percentage share of each component. This shows that the largest component of household consumption is food.

Consumption On	Description	% share in Total
Food		48.7
Power and Fuel	Cooking fuel, petrol, diesel, electricity	9.6
Cosmetics	Includes toiletries	7.1
Education	Books and various fees	5.5
Miscellaneous	Includes tourism, social obligations	4.8
Communication	Telephone, newspaper, TV, internet	4.6
Clothing	Garments, footwear and accessories	4.5
Transport	Bus/train/autorickshaw	3.8
Intoxicants	Cigarettes and alcohol	2.7
Rent	House rent and other charges	2.2
Monthly repayments (EMIs)	Installments on cars, durable goods, home	1.9
Restaurants		1.8
Health	Medicines, doctor fees, hospitalisations	1.6
Recreation	CDs, movies, toys	1.0

3.2 Consumption

Table 2 reports the various items that the average household in the sample surveyed in the database consumes. The largest component is food at almost 50% of expenditure. This along with expenditure on power items (including electricity, cooking fuel and petrol), cosmetics (which includes soaps and other toiletries), and education add to around three-quarters of the consumption expenditure of the average household in the sample.

3.3 Household borrowings

Household borrowing is presented as a binary variable in the CP database: whether households borrowed from a particular source. This is then calculated for all sources of borrowing. Table 3 presents the fraction of the sample that have borrowings. This is presented separately for different income categories as well as from different sources of credit, both informal (friends and family, money lenders) and formal (banks and SHG/MFIs). We extrapolate from the database to estimate that there are 28 million customers of micro-

credit (including both SHG and MFI) in India. (Srinivasan, 2012) separately estimated this as 27 million, validating the CP database as being consistent with other research in this area.

Table 3 shows that all income groups in India borrow from all sources. What varies is the extent to which a given lender is the dominant source of credit for different income groups. A higher fraction of the high income group take loans from banks, while a larger fraction of the lower income group use the micro-finance channels. The table also shows that the incidence of borrowing in AP is higher than that of the overall sample for India.

Table 4 presents the purpose for borrowing across income categories. This shows that consumption as the most important reason for borrowing, for all but one income category (I-3). A significant fraction of the higher income categories (I-3 to I-6) borrow to make investments and to purchase consumer durables, in contrast to the lower income groups.

4 Research design

The data described in the previous section allows us to compare the changes in average consumption before and after the ban for the homogenous regions (or HRs) in AP. It also allows us to identify HRs in the rest of India that are similar to the AP HRs, but where the ban was not implemented. This allows us to implement a DID estimation with the treatment group as the AP observations, and the counterfactuals which are HRs in states that did not ban micro-finance, but which match the treatment HRs on some observables and become the control group.

In this design, the first step is the procedure to identify these matched HRs using the procedure described in the following Section.

4.1 Matching procedure

In order to pair treatment and control HRs such that consumption in both regions is consistently comparable before the micro-finance ban, a set of observable characteristics are selected based on which these HRs can be matched. This is called the set of match covariates. A distance measure is then chosen using which to find the best match for each AP HR among the non-AP HRs.

Table 3 Who borrows and from where: India and AP in 2010-11

The table reports the fraction of the sample that borrow from various sources in the four quarters of fiscal year 2010-11, for India and for AP. The last column in the table shows the fraction of households in each income group in the region who have some borrowing. This varies from a fifth of the richest group to a bit less than half of the poorest in India. This changes dramatically in the case of AP where more than half the sample borrows. The table also shows the sources from which the sample borrows. Of the five sources, the SHG/MFI are the micro-finance lenders.

The role of banks peaks for the **I4** income group where the average annual income is Rs. 300,000 both in India and in AP. It steadily peters away when dealing with lower income groups, going down to less than 5% for the poorest in India, and less than 10% of the poorest in AP.

Micro-finance plays an important role for groups with an average annual income of less than Rs.100,000, serving around 8 % of the India sample and around 40% of the AP sample.

	No. of HH	Friends	Money lenders	Sources of Borrowing		Others	Any
				SHG/MFI	Banks		
India:							
I1 (Rich)	438	0.44	0.12	0.12	17.69	1.29	19.81
I2	887	2.92	2.86	0.28	15.29	3.24	19.14
I3	8222	7.88	6.56	0.70	18.77	6.85	27.08
I4	13200	9.87	7.71	1.65	19.17	7.30	31.53
I5	14314	11.60	7.83	2.09	14.59	7.23	32.42
I6	24434	16.24	10.23	4.05	13.09	8.69	36.52
I7	15189	20.60	13.01	6.12	10.41	10.66	41.97
I8	33389	21.65	14.47	6.99	7.31	11.28	40.27
I9	28796	24.69	14.26	7.64	5.20	11.67	42.55
I10	7786	29.46	14.06	7.06	4.60	10.11	46.07
I11 (Poor)	2658	30.00	13.33	6.65	3.13	8.71	44.91
Total	149313	20.42	12.45	5.76	9.18	10.10	36.69
AP:							
I1 (Rich)	2						
I2	6		67.76				67.76
I3	131	33.90	23.57	3.80	25.61	8.83	47.06
I4	471	30.72	21.45	20.58	26.10	8.51	55.67
I5	775	44.56	30.07	16.26	20.16	13.33	63.05
I6	2014	52.25	41.23	26.65	20.84	16.48	74.14
I7	1531	61.14	50.84	32.65	16.92	22.57	82.56
I8	3598	58.67	54.40	35.51	14.47	25.82	85.43
I9	2179	62.00	54.91	39.04	12.56	31.54	86.69
I10	200	60.00	58.28	34.06	9.14	29.32	88.02
I11 (Poor)	44	51.70	60.28	41.21	6.87	22.75	87.53
Total	10951	57.46	50.31	33.37	15.64	24.49	82.67

Table 4 Purpose of borrowing in AP, March 2010

This table presents the proportion of borrowers in AP that have borrowed for various purposes. For example, 69% of borrowing in the I-3 income group reported having borrowed for the purpose of housing. There could be multiple reasons to borrow. Among the columns in the table, *Number* is the number of borrower households in each income group; *Consumption* includes reasons of general consumption, but excludes reasons of health, marriage and education; *Investment* includes borrowing for business purposes as well as investments in other instruments.

	Number of households	Housing	Consumption	Durables	Investment	Debt repayment
I-3	46	45.2	46.8	10.8	55.7	27.3
I-4	192	29.8	51.7	7.6	53.1	22.0
I-5	389	36.1	60.3	15.5	56.8	27.2
I-6	1277	33.4	65.9	18.4	44.6	28.6
I-7	1127	33.4	72.2	20.8	36.8	28.9
I-8	2930	34.3	77.8	21.2	27.3	28.2
I-9	1907	32.6	80.6	21.1	20.0	30.1
I-10	175	28.7	77.9	13.2	21.4	27.5
I-11	37	32.2	70.3	16.2	24.8	39.2

We exclude the neighbouring states in South India from the control group. They may have suffered from an indirect treatment effect, either because of transmission of shocks through trade, or because of political actions by state governments that are similar to those undertaken in AP. These include the states of Tamil Nadu, Kerala and Karnataka.

Match covariates This contains two categories of characteristics: those that measure prosperity, and those that measure access to finance. The following are the chosen covariates:

1. Average household income.
2. Number of households.
3. Working population measured as the proportion of the HR that is in the working age. The working age is between 20 and 60 years.
4. Proportion of households who have graduated the 10th grade.
5. Proportion of the population that is financially excluded. This is measured as the fraction of households in the HR that do not have a bank account, credit card, life insurance policy, or other similar formal financial products.
6. Proportion of farmers in the region.⁷

⁷The household participation in the agricultural sector of a region has implications for

We do not use the presence of micro-finance institutions as a match variable because it is directly effected by the treatment. We also do not use the observed outcome variables in the matching procedure as it introduces certain difficulties in estimation (Stuart, 2010).

Distance measure We use the Mahalonobis distance measure for nearest neighbour matching. The measure is calculated as:

$$D_{ij} = [(X_i - X_j)' \Sigma^{-1} (X_i - X_j)]^{\frac{1}{2}}$$

where D_{ij} is the distance between unit i and j and X_i and X_j are the characteristics of the control and treatment units. A “1:1 nearest neighbor matching” method selects the control unit with the smallest distance from any given treated unit i .

This measure was used because the policy change was exogenous. Alternative measures are tested as part of the sensitivity analysis in Section 6.2.

4.2 The difference-in-difference (DID) estimator

Once the treatment and control HRs are identified, the following DID model is estimated:

$$C_{i,t} = \beta_0 + \beta_1 AP_{i,t} + \beta_2 POST-CRISIS_{i,t} + \beta_3 (AP_{i,t} \times POST-CRISIS_{i,t}) + \epsilon_{i,t}$$

where $C_{i,t}$ is the average household consumption, AP is a dummy which takes value “1” if i is a region in AP (the treatment HR) and “0” otherwise (the control HR). $POST-CRISIS$ captures whether the observation is from the period before the micro-finance ban (post-crisis = “0”) or after (post-crisis = “1”). $PRE-CRISIS$ quarters include the four quarters of March to December 2010, and $POST-CRISIS$ includes the four quarters of March to December 2011. $\hat{\beta}_3$ will be negative and statistically significant if the average consumption in the treatment HRs sees a greater fall after the ban compared to the matched HRs over the same time period.

The matching DID estimator is a considerable improvement on standard matching estimators (Blundell and Dias, 2000). It eliminates unobserved, time-invariant differences between the treatment and control groups that standard matching estimators fail to eliminate (Smith and Todd, 2005). It is also an improvement on a simple DID where the treatment and control units may not have match balance.

fluctuations in income and therefore, the need to access finance for consumption smoothing (Rosenzweig, 2001).

Table 5 Match balance using t-stat and standardised difference

This table presents the match balance statistics between the treatment and control group. The p values (p-val) are generated from the t-test, and SDIFF reflects the standardised difference.

% balance improvement refers to the improvement in balance after matching for all the covariates.

	Means Treated	Means Control	Mean Diff	t-stat	p-val	SDIFF	% Bal. Impr.
Average HH income	10.25	10.33	-0.09	-0.80	0.43	-24.78	58.30
No.of HH	7.08	6.71	0.37	1.58	0.13	40.66	52.99
Working	3.65	3.47	0.17	0.29	0.77	7.16	49.08
Graduated 10 th grade	2.59	2.51	0.08	-0.52	0.60	-12.21	84.65
Financial excluded	4.35	4.27	0.08	1.24	0.23	28.49	91.71
Farmer	0.36	0.24	0.12	-0.84	0.41	-19.69	80.01

4.3 Evaluating the design

We now evaluate the design from three points of view. Is there match balance? How does the design fare when presented with a placebo (a non-event)? How much power does the design have?

4.3.1 Is there match balance?

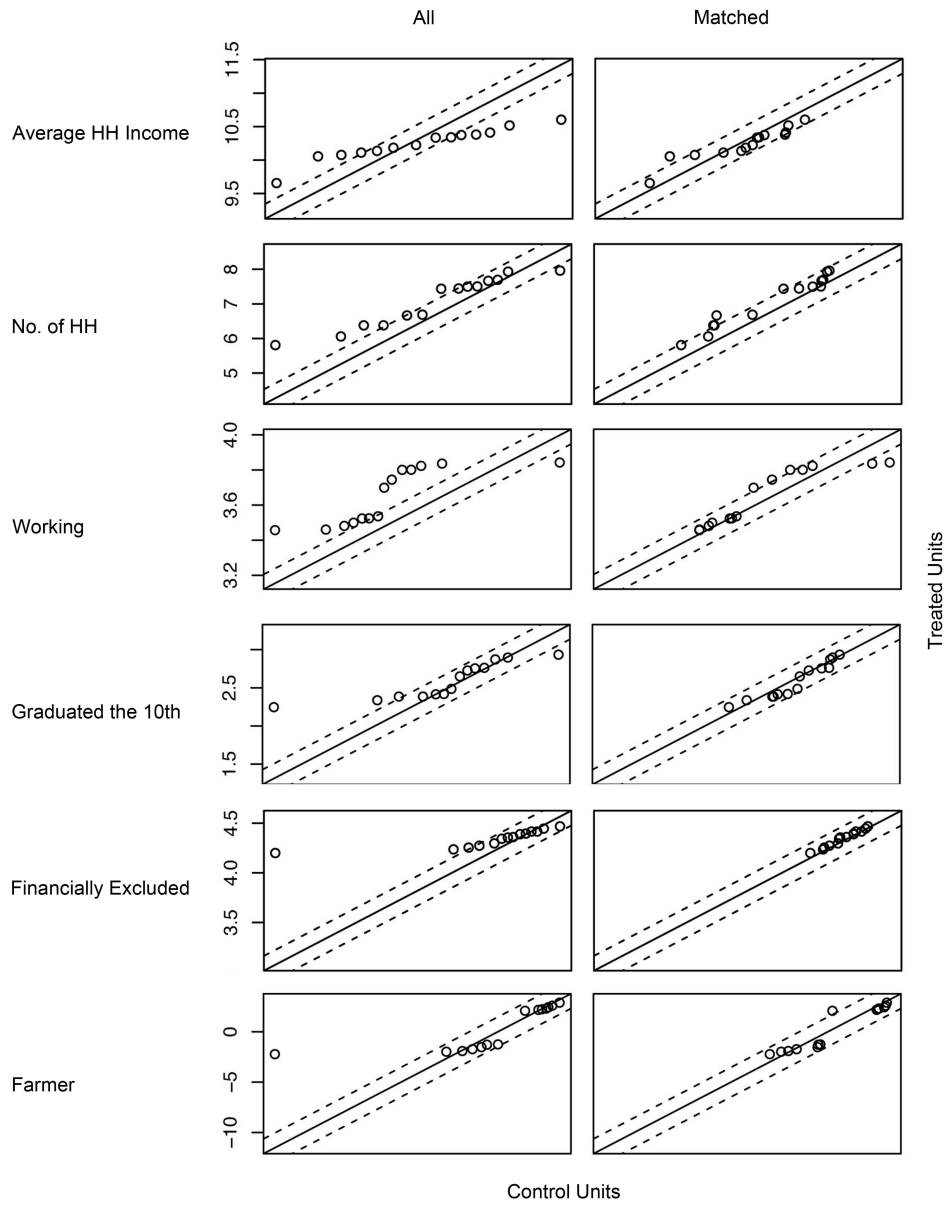
In a sound design, match balance should be achieved between the treatment and control HRs. This implies that the distribution of the X co-variables observed for the treatment and control HRs are equal in the pre-treatment period. When match balance is inadequate, the estimation of treatment effects relies on extrapolation, which is fraught with difficulties (Rosenbaum and Rubin, 1983).

Stuart (2010) points out that the parametric tests such as the t-test are often not accurate measures of balance as they compare only the averages, while the entire distribution matters in a matching exercise. One test for this is a visual inspection of the quantile-quantile (QQ) plots for each covariate pair used in the matching exercise. If there is match balance, the points would fall on the 45° line. Figure 1 shows that this is not the case with the full dataset, but is substantially the case with the matched data.

A final test is the Hotelling's T-square test which considers the joint significance of the differences between the co-variables. In the sample, the χ -statistic is 10.44 with a p-value of 0.12 and the F-statistic is 1.35 with a p-value of

Figure 1 QQ plots of the match covariates

The figure plots the QQ-plots of the covariates used in matching, before and after the matching exercise. The y-axis in each box reflects the treated units and the x-axis the control units. Deviations from the 45° line indicate differences in the empirical distribution and a low match balance. In all cases, there is good match balance after the matching is done.



0.27. Both these indicate that the differences between the covariates are not jointly significant.

All four tests confirm that match balance has been achieved, and that we have a sound design.

4.3.2 Analysing a placebo

One test of the soundness of the estimation strategy is how it behaves in a Monte-Carlo simulation with a placebo. 14 HRs are selected at random as the treatment group where the AP HRs are excluded. For the selected HRs, we replicate the matching exercise to find control HRs, and test the difference between this treatment and control HRs using a DID estimation.

This simulation is repeated 10,000 times, and we find that the null of no change is rejected 0% of the time. While the rejection rate is not exactly the same as the size of the test, these results serves as a check that the estimation strategy does not reject the null incorrectly when presented with a placebo.

Another potential concern of the matching exercise is that the consumption between AP and the matched HRs was different even before the micro-finance ban. We test another placebo, where the treatment and control HRs are as presented in Section 4.1 but using the periods prior to the micro-finance ban from 2009 upto the last quarter of 2010. If there is no significant difference in the average consumption between the AP and matched HRs in the periods prior to the ban, it is likely that the results are not driven by differences in prior consumption. These gives us $\hat{\beta}_3 = -1815.4$, with s.e. = 1259.56 and t-test = -1.44 . This shows that the difference in average consumption in AP and control HRs between 2009 and 2010 is not statistically significant.

4.3.3 What is the size of the impact that can be resolved by the design?

Since the data only has observations for 14 HRs in AP where the ban applies out of 200, our ability to reject the null, when the null is not true, may be limited owing to small sample. Hence, we undertake a power study in order to assess the magnitude of deviations from the null that can be detected through our design and estimation strategy.

This is implemented using simulations where 14 HRs are randomly chosen to be a treatment group, and a shock is artificially induced into this set. The

matching exercise is repeated and a DID estimator is used to test whether there is a statistically significant difference between the treatment and the control at 95% level of significance. We find that the size of the change in consumption that can be discerned with a 70% probability in Rs.6000. This implies that a change in total consumption of Rs.6000 per quarter is discerned by a 95 percent test with 70% probability. Conversely, if our analysis showed that the null (no effect) cannot be rejected, this would imply that the true effect size is smaller than Rs.6000. This shows that the research design has fairly high power.

5 Results

5.1 Did consumption fall when access to micro-credit was reduced?

Figure 2 plots the average household consumption in the 14 AP HRs as the solid line, in the graph from March 2010 upto September 2011. The average household consumption for the control HRs is plotted as the dashed line.

The impact of the ban on micro-finance in AP is measured by the change in average household consumption in the AP HRs, before and after the ban relative to the change in the control HRs. Figure 2 compares the average consumption one quarter before the ban (Dec 2010) and one quarter after the ban (June 2011). The graph shows that average consumption of AP households *dropped* between the pre-ban and post-ban quarters, while it *rose* for the non-AP households.

The average AP household consumed Rs.17,531 in a quarter in the pre-ban period. This consumption dropped to Rs.15,487 in the post-ban period, which is *a decrease of 12 percent* between two quarters. In contrast, the average non-AP households consumed Rs.16,574 in the pre-ban period and Rs.17,727 in the post-ban period, which is *an increase of seven percent*. The net average drop in quarterly consumption in AP was 19 percent, as a consequence of the micro-finance ban.

We also estimate the impact of the ban using the estimate of β_3 from the DID models proposed in Section 4.2. Table 6 shows $\hat{\beta}_3 = -3375.1$, significant at 95 percent level of confidence. This implies an adverse impact on consumption in the AP households of 19.5% due to the micro-finance ban.⁸ This is consistent

⁸DID regressions on components of total expenditure such as food, power and fuel,

Figure 2 Average quarterly household consumption, March 2010-September 2011

The graph shows the average total household consumption in the treatment HRs (in AP) and control HRs (matched, outside AP) between the March 2010 to September 2011 quarters.

The vertical bar marks the quarter in which the state government passed the micro-finance ban in Dec 2010. While consumption in the controls grew steadily through this period, consumption in AP dropped sharply from Rs.17,530 to Rs.15,190.

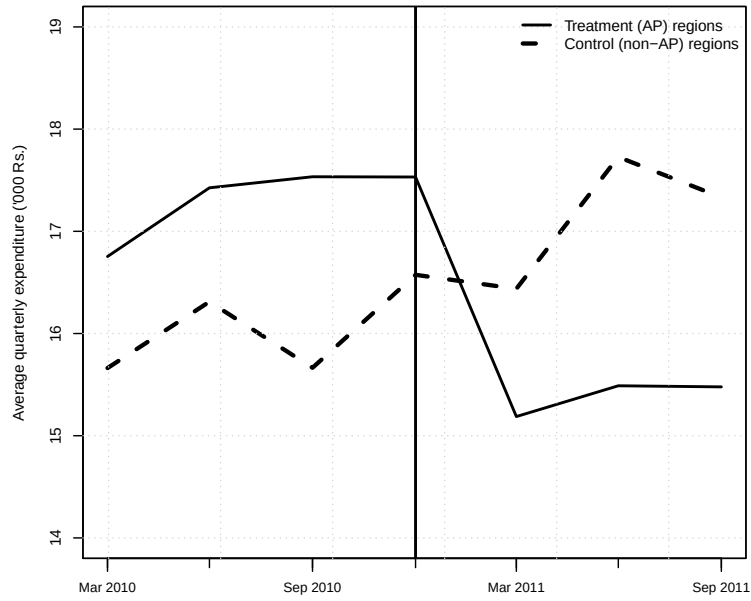


Table 6 DID estimates for average household consumption across treatment and control HRs

The table presents $\hat{\beta}_3$ from the following DID estimation:

$$C_{i,t} = \beta_0 + \beta_1 AP_{i,t} + \beta_2 POST-CRISIS_{i,t} + \beta_3 (AP_{i,t} \times POST-CRISIS_{i,t}) + \epsilon_{i,t}$$

The table also reports heteroskedasticity-consistent standard errors (std.err.), p-values (p.val), the adjusted p-values (adj.p) and Q_{power} which is the effect size for different consumption components that could be detected with 70 percent probability.

$\hat{\beta}_3$	std.err.	p.val	Adj.p	Q_{power}
-3375.1	1450.5	0.02	0.05**	6000

** indicates significance at 5%

with the 19% drop calculated from Figure 2.

Together, the above evidence leads to the conclusion that the micro-finance ban implemented by the State Government caused a significant adverse impact on the overall average household consumption in AP. This result applies to the average household, without differentiating between the various income categories.

Does the extent of indebtedness in AP matter?

In the matching exercise, we explicitly ignored all aspects of household characteristics that were related to lending, and instead focussed on pure economic characteristics to identify matched homogenous regions (HRs). However, Section 3.3 shows the average household in AP had a higher level of loans compared to the average level in India. One of the main motivations used by the AP State Government in implementing the ban with rapidity was that build up of household debt beyond their ability to repay would inevitably lead to households entering financial distress at a later point in time. This raises the question of whether the intervention shows a negative impact because the matching covariates capture nothing about the level of borrowings of the treatment and control groups. In comparison to control HRs where the households have as high a level of borrowings, would a DID estimation reveal that the AP HRs are better off after the micro-credit ban?

In order to test this, we carry out two searches for new controls that better matches the indebtedness of the AP HRs:

1. Add the fraction of household with borrowings as a match covariate to the existing six covariates.

We identify a fresh set of control HRs and re-estimate the DID regression.⁹ If the high indebtedness of AP HRs does lead to a reduction of household welfare, then the drop in consumption seen in AP after the ban will be much less than a drop seen in the other HRs that are similarly indebted.

2. Add the states of South India as candidates for the control group.¹⁰

cosmetics and education, also showed negative and statistically significant effects. These are available upon request.

⁹We continue to achieve match balance on the initial variables. The match balance on the proportion of indebtedness households was mixed: the standardized difference was 67.9, and the t-stat was statistically significant at 5%. However, the KS-stat was at 0.34 indicating that the distributions were relatively close even if the mean was not.

¹⁰We achieve match balance on all variables.

Table 7 DID estimates adjusting for the high indebtedness in AP

The table presents $\hat{\beta}_3$ from the following DID estimation:

$$C_{i,t} = \beta_0 + \beta_1 \text{AP}_{i,t} + \beta_2 \text{POST-CRISIS}_{i,t} + \beta_3 (\text{AP}_{i,t} \times \text{POST-CRISIS}_{i,t}) + \epsilon_{i,t}$$

The table also reports heteroskedasticity-consistent standard errors (std.err.), p-values (p.val), the adjusted p-values (adj.p) and Q_{power} which is the effect size for different consumption components that could be detected with 70 percent probability.

Adjustments to matching	$\hat{\beta}_3$	std.err.	p.val	Adj.p
Include fraction of borrower households	-2983.77	1411.94	0.04	0.08*
Include South Indian states	-3069.82	1409.66	0.03	0.08*

* indicates significance at 10%

Section 4.1 argued that HRs from the states of South India were removed from the control pool in order to avoid spillover effects. However, these states have similar levels of indebtedness as AP, and would capture the indebtedness factor if selected into the control group. To the extent that a certain partial treatment effect was probably found all over South India, the use of HRs from South India in the control pool would generate a downward bias in the measured treatment effect.

We re-run the matching exercise to identify a fresh set of control HRs including the southern states, and re-estimate the DID regression.

Table 7 presents the results of these two DID re-estimations that adjust for the high indebtedness of AP HRs. In case of both the new set of controls, the new DID estimates for $\hat{\beta}_3$ are around -3000, slightly smaller in magnitude compared to Rs.3375 in Table 6. The significance of the coefficients have also dropped slightly to around 90% (as opposed to 95% in the previous estimation). This implies a drop in average household consumption with the adjustment for indebtedness of AP of 17.33% as a consequence of the micro-finance ban, only slightly lower than the 19.5% estimated previously.

Thus, even after having designed the estimation approach to account for the high indebtedness of AP households, the evidence points to a sharp and negative impact on average household consumption because of the ban.

Table 8 Volatility of consumption before and after the ban

The table presents the volatility of average household consumption for the AP and the control HRs, in the pre-ban and the post-ban quarters, where * indicates that $\sigma_{\text{post-ban}}$ is higher at 94% level of significance.

	Treatment HRs		Control HRs	
	$\sigma_{\text{pre-ban}}$	$\sigma_{\text{post-ban}}$	$\sigma_{\text{pre-ban}}$	$\sigma_{\text{post-ban}}$
Average household consumption	2.12	8.89*	5.19	5.29

5.2 Did volatility of consumption rise with lower access to micro-finance?

Since access to finance is important for consumption smoothing, the withdrawal of such access could impact not just on the level of consumption but also the volatility of consumption. Thus, we expect higher volatility of this average consumption, while we would not expect to see any change in the volatility of consumption in the control HRs.

Table 8 presents the standard deviation of the percentage change in average household expenditure in the period before the ban as $\sigma_{\text{pre-ban}}$ and after the ban as $\sigma_{\text{post-ban}}$. We calculate the percentage change in the average expenditure for the 14 HRs in AP and the 14 control HRs as seen in Figure 2. The table suggests that there has been an increase in the volatility of household expenditure between the treatment and control HRs. $\sigma_{\text{post-ban}}$ for consumption expenditure is higher for the AP HRs, while it is not significantly higher for the control.¹¹

5.3 Is there an effect on average saving?

The credit withdrawal in AP came with a one-time windfall gain at the time of the ban since borrowers were encouraged to default en-masse on their

¹¹The impact of the ban on volatility of consumption expenditure was analysed using a variety of econometric approaches. A simple comparison of the pre- and post-ban periods volatility was compared using the average of the volatility for each of the 14 treatment HRs and their matched controls. Here also, the volatility of average consumption showed an increase, though at a 90% level of significance. Next, the impact of the ban was also analysed using the DID framework. The estimated coefficient on the interaction term between AP dummy and the post-crisis dummy was consistently positive across all these approaches showing an increase in the volatility after the ban. The details of these alternative estimations can be made available on request.

loans by local politicians. However, the windfall may not be beneficial for two reasons. First, the value of the windfall at a particular point in time is the value of the repayment due at the time of the windfall. This may be a small or a large amount depending on where the household was in its credit cycle. Zeldes (1997) suggested that even if the current liquidity constraints are not binding, the possibility that such constraints will bind in the future will lower the current consumption of a risk-averse individual. Further, higher uncertainty leads households to hold higher buffer-stocks, leading to declines in expenditure on consumption.

Table 9 compares the average household savings and the saving rate for treatment and control HRs, and finds that the saving rate is higher for the treatment HRs. These results suggest that the households in the treatment HRs are building up buffer stocks to hedge against future uncertainty, and a future lack of credit access.

Table 9 DID on average household savings and savings rate

The tables shows the DID estimates of average savings and savings rates in the treatment and control groups.

	coeff	std.error.HC	p.val.HC	adj.p.val
Average hh. savings	-449.59	1709.69	0.79	0.79
Saving rate	7.03	3.55	0.05	0.10*

5.4 Which groups were more affected?

The CP database allows us to compare expenditures in rural HRs compared to urban HRs. We expect that rural regions will suffer a higher impact as a consequence of the micro-finance ban compared with urban regions owing to greater dependence of micro-finance in the former.

In order to test the effect of the ban, we estimate a DID model with a slight modification of the model in Section 4.2 as follows:

$$C_{i,t} = \beta_0 + \beta_1 AP_{i,t} + \beta_2 POST-CRISIS_{i,t} + \beta_3 (AP_{i,t} \times POST-CRISIS_{i,t}) + \beta_4 (AP_{i,t} \times POST-CRISIS_{i,t} \times RURAL_{i,t}) + \epsilon_{i,t}$$

where $RURAL_{i,t}$ takes the value 1 if the i^{th} HR is rural and 0 if it is urban. Then β_4 captures the effect of the ban on a rural HR in the treatment units compared with a rural HR in the control unit. The estimated results are presented in Table 10, and show that consumption fell in rural AP for average

Table 10 DID estimates between rural and urban HRs

This table presents the estimation results for:

$$C_{i,t} = \beta_0 + \beta_1 AP_{i,t} + \beta_2 POST-CRISIS_{i,t} + \beta_3 (AP_{i,t} \times POST-CRISIS_{i,t}) + \beta_4 (AP_{i,t} \times POST-CRISIS_{i,t} \times RURAL_{i,t}) + \epsilon_{i,t}$$

where $RURAL_{i,t}$ takes the value 1 if the i^{th} HR is rural and 0 if it is urban. β_4 is the impact of the ban on rural AP HRs compared to urban HRs. The results include the heteroskedasticity consistent standard errors and the adjusted p-values.

$\hat{\beta}_4$	std.error	adj.p
-5349.4	1175.8	0.00***

*** indicates significance at 99%

consumption. This suggests that regions lacking wider access to finance suffered more as a consequence of the ban.

The results are based on aggregates across households (average values of consumption) rather than individual households. Given the lack of household level data, it is difficult to establish cross-sectional variation in how households with different liquidity constraints respond to the impact of the credit ban. What these results do suggest is that urban households, that can access more credit sources, are less vulnerable to shocks in their consumption, compared to those households with fewer sources of credit access. The shocks to consumption are long-lasting. These results show that consumption had not recovered four quarters after the crisis, implying that once access to credit is taken away, it is very difficult for other sources to compensate for it.¹²

6 Threats to validity

An assumption in the current research design is that outcomes from the treatment and control group are likely to be driven by a set of common factors, and that the only differing factor is the micro-finance ban. In that case, the fall in average consumption observed in the AP HRs can be solely attributed to the ban. In this section we address alternative explanations that might explain the fall in the average consumption. Three alternatives can be offered:

¹²We find that the consumption gap continued to be significant between the treatment and control HRs even in the first two quarters of 2012. These results are not reported here but can be made available on request.

1. Some other event in AP caused the same result.
2. The results are sensitive to the matching strategy.

6.1 Did another event in AP cause the results?

A possible criticism is that some event in AP, other than the micro-credit ban, is what caused the drop in consumption observed in Section 5. For instance, if there was a drought or a flood in AP but not the other states, this would adversely effect the average consumption of the AP HRs but not the controls.¹³

One way to address this concern is to analyse the drop in consumption in a given HR by the exposure of the HR to micro-finance. The exposure to micro-finance is the fraction of borrower households in an HR that have loans from SHG/MFI (Table 3). If a treated HR with a higher micro-finance exposure sees a larger drop in consumption after the ban, the drop can be more confidently attributed to the ban.

Figure 3 presents the change in average household expenditure for the 14 AP HRs by the micro-finance exposure of the HR. Consumption dropped for 13 out of 14 HRs. The decline in consumption is larger on average with higher exposure to micro-finance. We test this statistically by estimating the regression:

$$\text{Impact}_i = \alpha + \beta \times \text{Micro-finance exposure}_i + \epsilon_i$$

where we expect $\hat{\beta}$ to be negative and significant if the change is because of the micro-finance ban. A robust regression of the impact of the ban on the consumption of HR i on the micro-finance exposure of HR i yields $\hat{\beta} = -0.27$ with a t-statistic of -1.98 . This suggests that it was the ban that caused the drop in the consumption in the AP HRs, because other events would not cause this manner of cross-sectional variation in decreased consumption.

Another method of testing if other changes were driving the fall in consumption (such as the drought) is to examine changes in average income of HRs. If we do not find a significant fall in the average income of HRs, then we can more confidently claim that the consumption decline was a result of the ban.

Table 11 reports the DID regression of the average income in the treatment and control HRs. The estimated coefficient is negative but not significant. This implies that there was a likely negative shock on income in the treatment

¹³There was a drought in AP in the period of July to September 2011. However, this falls after the period analysed in this paper.

Figure 3 Impact of the ban by micro-finance exposure

This figure plots the impact of the ban on household consumption in the AP HR by the micro-finance exposure of that HR. Here, impact is measured as the change in consumption expenditure in the AP HR before and after the ban, and micro-finance exposure of the HR is measured as the fraction of borrower households in AP that have borrowed from the SHG/MFI category. The graph shows that there is a sharper fall in consumption with higher exposure to micro-finance.

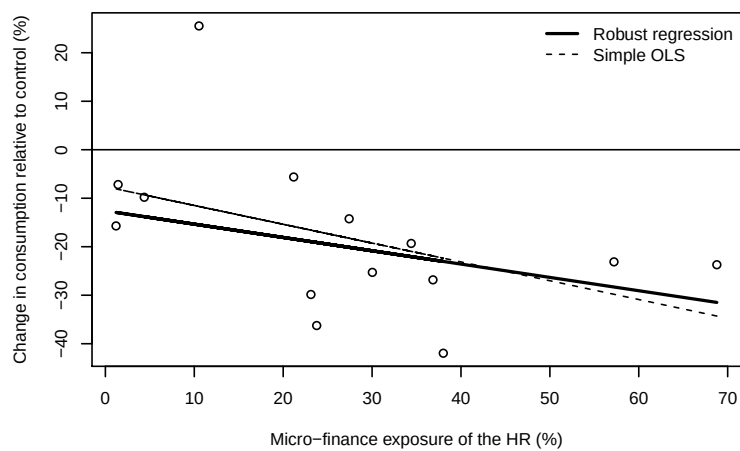


Table 11 DID estimates of changes in average income

This table presents the $\hat{\beta}_3$ DID estimators from:

$$AI_{i,t} = \beta_0 + \beta_1 AP_{i,t} + \beta_2 \text{POST-CRISIS}_{i,t} + \beta_3 (AP_{i,t} \times \text{POST-CRISIS}_{i,t}) + \epsilon_{i,t}$$

where AI is the average income of the HR. The results include the heteroskedasticity consistent standard errors and the adjusted p-values.

	$\hat{\beta}_3$	std.err	p.val	adj.p
Average income	-3522.6	2236.6	0.12	0.29

HRs after the ban, though not statistically significant. This suggests that there may have been some factor in AP that may have contributed to the drop in consumption due to the micro-finance ban. However, it is unlikely that it was sufficient to have caused the large and significant negative drop in consumption, particularly given that the drop in consumption across the regions is strongly correlated with the micro-finance exposure in the region.

6.2 Are the results sensitive to the matching strategy?

One concern is how sensitive the results are to either the set of matching covariates or the specific matching algorithm used for the results in Section 5. We re-estimate the DID using other matching approaches including:

1. Create a new set of matching covariates by dropping one covariate used in the original matching procedure, re-estimating the DID and testing for the change in results with the new matched HRs.
2. Using a *genetic matching* algorithm over the same covariates to obtain a set of control HRs.

This is a method of multivariate matching, that uses an evolutionary search algorithm to determine the weight each covariate is given (Diamond and Sekhon, 2012).

3. Using the “proportion of women in an HR” as a match variable as a proxy for exposure to micro-credit, since micro-finance institutions (either SHG or MFI) typically lend to women.
4. Using the “proportion of households with debt outstanding to SHG/MFI” as a match variable to capture the differences in the exposure to micro-credit.

Table 12 DID estimates across varying matching strategies

	$\hat{\beta}_3$	std.err	p.val	adj.p
Baseline result				
	-3375.1	1450.5	0.02	0.05**
1. Dropping one covariate at a time				
Average household income	-4100.30	1595.19	0.01	0.03**
Number of households	-3750.54	1414.87	0.01	0.03**
Working age population	-2862.41	1340.42	0.03	0.08*
Graduated 10 th grade	-3993.80	1452.54	0.01	0.02**
Financially excluded	-2794.62	1508.09	0.07	0.16
Farmer	-3556.64	1403.00	0.01	0.03**
2. Using a genetic matching algorithm				
	-4648.2	1834.2	0.01	0.04**
3. Adding proportion of women				
	-3105.20	1619.58	0.06	0.14
*** indicates 1%, ** 5% and * 10%				

Table 12 shows estimates from these DID estimations. All the estimated coefficients are negative, and all but one are significant. While different approaches yield coefficients that vary in magnitude, the direction and significance in each case is consistent with those of the estimates presented in Table 6. Thus, we infer that our original results are not sensitive to the matching strategy used.

7 Conclusion

The rapid growth in the micro-credit business by financial firms in recent decades have been dogged by concerns about consumer protection. Policy makers worldwide have implemented a variety of interventions, ranging from relatively subtle rules on consumer protection to restrictions. The optimal public policy response to new kinds of micro-credit firms is constantly evolving, drawing from a small literature analysing the impact of these interventions.

The contribution of this paper lies in utilising a large natural experiment – a complete ban on micro-finance in the Indian state of Andhra Pradesh (AP) that has the population of Germany – and in having high quality measurement of average consumption from a large panel of households observed every quarter all over the country. Since the ban was imposed in only one state, we draw controls from other locations in the country to draw causal

inference about the impact.

The results suggest a fairly large negative impact of the ban on micro-finance. Consumption dropped by 17.3% as a consequence, with the drop being even higher in rural regions where the choices to access formal sources of finance is lower, and volatility of consumption increased. The results also suggest that households in AP may have increased savings as an attempt to hedge against the uncertainty of future credit access. Thus, while the ban on the micro-finance industry was initiated by policy makers in AP under the claim that this would help poor people, in reality it has imposed real costs on everyone in the economy.

There are intriguing analogies between the experiment in AP – where a subset of society that was using micro-finance abruptly lost credit access – and the macroeconomics and finance literature on deleveraging (Eggertsson and Krugman, 2012). This literature focuses on what happens when some borrowers in a country are highly indebted and face an abrupt shock to credit access. The natural experiment that we have examined in AP appears to have similar characteristics. Micro-credit is typically a small component of the larger credit outstanding by value. An assumption is that disruptions to the micro-credit industry would impact only those who have borrowed from these firms. However, a large scale withdrawal of micro-credit may have larger consequences where access to capital is generally constrained.

A drawback of the analysis is that we only observe household aggregates at the level of geographical areas and income classes, rather than individual households. Record level data might reveal that welfare is improved without micro-finance, for *certain* households. Since we lack household level records, we cannot distinguish between a bigger impact on the households that directly used micro finance, and indirect effects on their peers in the same income class. When such data is eventually released, these effects could be measured.

The findings in this paper suggest that the overall average treatment effect associated with banning micro-finance in Andhra Pradesh was negative. One lesson is that a blunt policy instrument, such as a complete ban of micro-finance, is inadvisable. In the global debate about the welfare consequences of for-profit micro finance, this would suggest that extreme government restrictions are ill-advised. Thus, even though this analysis does not rule out the potential presence of market failure in the form of weak decision making by some poor people, the optimal response to market failures involves a more subtle approach of consumer protection, rather than the blunt instrument of a ban. These questions are important avenues for future research.

References

- Arunachalam, Ramesh S. (2010): “The 2010 Andhra Pradesh Micro-Finance Crisis Revisited...,” Blog: ”Candid Unheard Voice of Indian Micro-finance”, <http://microfinance-in-india.blogspot.com/2010/12/2010-andhra-pradesh-micro-finance.html>.
- Augsburg et. al. (2012): “Microfinance, poverty and education,” Working Papers 18538, NBER.
- Banerjee, Abhijit V., Esther Duflo, Rachel Glennerster, and Cynthia Kinnan (2013): “The Miracle of Microfinance? Evidence from a randomized evaluation,” Working Paper 13-09, MIT Department of Economics.
- Blundell, Richard, and Monica Costa Dias (2000): “Evaluation methods for non-experimental data,” *Fiscal Studies*, 21(4), 427–468.
- Campbell, John Y. (2006): “Household finance,” *The Journal of Finance*, 60(4).
- Crépon, Bruno, Florencia Devoto, Esther Duflo, and William Parienté (2011): “Impact of microcredit in rural areas of Morocco: Evidence from a Randomized Evaluation,” .
- Datta, Sankar, and Vijay Mahajan (2003): “Savings and Credit Movement of Andhra Pradesh,” Discussion paper, BASIX.
- Diamond, Alexis, and Jasjeet S. Sekhon (2012): “Genetic Matching for Estimating Causal Effects: A General Multivariate Matching Method for Achieving Balance in Observational Studies,” *Review of Economics and Statistics*, (forthcoming).
- Eggertsson, Gauti B., and Paul Krugman (2012): “Debt, Deleveraging and the Liquidity Trap: A Fisher-Minsky-Koo Approach,” *Quarterly Journal of Economics*, 127(3), 1469–1513.
- Kaboski, Joseph P., and Robert M. Townsend (2012): “The Impacts of Credit on Village Economies,” *American Economic Journal: Applied Economics*, 4(2), 98–133.
- Karlan, Dean, and Jonathan Zinman (2010): “Expanding Credit Access: Using Randomized Supply Decisions to Estimate the Impacts,” *Review of Financial Studies*, 23(1), 433–464.
- (2011): “Microcredit in Theory and Practice: Using Randomized Credit Scoring for Impact Evaluation,” *Science*, 332, 1278–1284.

- Melzer, Brian T. (2011): “The real costs of credit access: Evidence from the payday lending market,” *The Quarterly Journal of Economics*, 126, 517–555.
- MFIN (2012): “The MicroScape,” Discussion paper, Microfinance Institutions Network.
- Mian, Atif, and Amir Sufi (2010): “Household Leverage and the Recession of 200709,” *IMF Economic Review*, 58(1), 74–117.
- Morgan, Donald P., and Michael Strain (2008): “Payday Holiday: How Households Fare After Payday Credit Bans,” Staff Report 309, Federal Reserve Bank of New York.
- Morse, Adair (2011): “Payday lenders: Heroes or villains?,” *Journal of Financial Economics*, 102(1), 28–44.
- Rosenbaum, Paul R., and Donald R. Rubin (1983): “The central role of the propensity score in observational studies for causal effects,” *Biometrika*, 70, 41–55.
- Rosenzweig, Mark R. (2001): “Savings Behaviour in Low-Income Countries,” *Oxford Review of Economic Policy*, 17(1), 40–54.
- Sa-dhan (2007): “Core Values and Voluntary Mutual Code of Conduct For Microfinance Institutions,” Discussion paper, Sa-Dhan : The Association of Community Development Finance Institutions.
- Sane, Renuka, and Susan Thomas (2013): “Regulating microfinance institutions,” *Economic and Political Weekly*, 68(5).
- Shylendra, H. S. (2006): “Microfinance Institutions in Andhra Pradesh,” *Economic and Political Weekly*, pp. 1959–1963.
- Smith, Jeffrey A., and Petra E. Todd (2005): “Does matching overcome LaLonde’s critique of nonexperimental estimators,” *Journal of Econometrics*, 125, 305–353.
- Srinivasan, N. (2010): “Microfinance: State of the Sector Report 2010,” Discussion paper, Access Development Services, Sage Publications.
- (2012): “Microfinance India: State of the Sector Report 2011,” Discussion paper, Access Development Services, Sage Publications.
- State government of Andhra Pradesh (2010): “An Ordinance to protect the women Self Help Groups from exploitation by the Micro Finance Insti-

- tutions in the State of Andhra Pradesh and for the matters connected therewith or incidental thereto,” State government of Andhra Pradesh.
- Stuart, Elizabeth A. (2010): “Matching methods for causal inference: A review and a look,” *Stat Sci*, 25(1), 1–21.
- Yerramilli, Pooja (2013): “The Politics Of The Microfinance Crisis In Andhra Pradesh, India,” *Journal of Politics & Society*, pp. 190–225.
- Zeldes, Stephen P. (1997): “Consumption and Liquidity Constraints: An Empirical Investigation,” *Journal of Political Economy*, 97(2), 305–346.
- Zinman, Jonathan (2010): “Restricting consumer credit access: Household survey evidence on effects around the Oregon rate cap,” *Journal of Banking and Finance*, 34(3), 546–556.

Appendix

A The CES data

The standard data-set on consumption expenditure in India is the quinquennial survey, the Consumption Expenditure Survey (CES), carried out by the National Sample Survey Organisation. The last two surveys were carried out in 2009-2010 (known as the 66th round) and 2011-12 (known as the 68th round).¹⁴ These years make the CES suitable for the study of the impact of the Andhra Pradesh ban on consumption expenditure. There are however three reasons that the NSS data was not used for this analysis. These are as follows:

- The CES is not a panel-survey. This makes it difficult to compare consumption in households across the two time-periods.
- The 66th and 68th rounds of the CES are two years apart. This makes it difficult to study consumption expenditures immediately before and after the ban.
- While the NSS has several sub-periods within a year, the sample in a state gets divided across the sub-periods, leading to very small sample sizes for each sub-period, even for aggregates. For example, the sample in Andhra Pradesh in CP every quarter is about 7500 households. The total Andhra Pradesh sample in NSSO is 6899.¹⁵
- The CES uses a moving reference period i.e. it asks households to report expenditure over the past 7 (or 30) days preceding the date of the survey. This makes it difficult to compare outcomes across households. CP on the other hand, uses a fixed reference period, that of the last three months.

B List of districts in the control

¹⁴The 68th round was carried out by a decision of the National Statistical Commission, even though it was only two years after the 66th round.

¹⁵Table 2.2, Key Indicators of Household Consumer Expenditure in India, 2011-2012

Table B.1 List of districts that make up the control

The table lists the names of all the districts that are covered in the HRs that make up the control in the paper.

Districts in Bihar: Paschim Champaran Purba Champaran Sheohar Sitamarhi Madhubani Saharsa Patna Bhojpur Buxar Kaimur (Bhabua) Rohtas Jehanabad Aurangabad Districts in Gujarat: Banas Kantha Patan Mahesana Sabar Kantha Panch Mahals Dohad Vadodara Narmada Bharuch Surat The Dangs Navsari Valsad Kachchh Jamnagar Sundernagar Rajkot Porbandar Junagadh Amreli Bhavnagar	Districts in Odisha: Bargarh Jharsuguda Sambalpur Debagarh Sundargarh Anugul Kandhmal Sonapur Districts in Madhya Pradesh: Jhabua West Nimar Barwani East Nimar Umaria Jabalpur Narsimhapur Dindori Mandla Chhindwara Seoni Balaghat Districts in Maharashtra: Washim Bhandara Gondiya Gadchiroli Chandrapur Yavatmal Hingoli Districts in West Bengal: Darjiling Jalpaiguri Koch Bihar
--	--
